CLUB club

2026-10-16

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Today we'll cover how to make a multiple sequence alignment.

Context

You've made consensus sequences from your mapped FASTQ alignment (SAM files) and you intend to make a phylogenetic tree.
You've downloaded some reference files from NCBI that you want to compare your consensus(es) to.

What multiple sequence alignment does

Simplistically, MSA figures out a way to insert gaps into sequences to make nucleotides or amino acids align column-wise as well as possible.



Once again, this is actually very complicated (assuming you try to do a good job).

There are many programs to make an MSA

Some of the main ones:

- MAFFT
- MUSCLE 5
- Clustal Omega
- FAMSA2
- T-Coffee
- PRANK

Let's start by doing something

Today we'll play with MAFFT. This is probably the most-widely used MSA tool.

It has 170 command-line options. Unfortunately (and inexplicably) the description of these are spread over various web pages. It can be hard to track them down or even to know they exist.

Install MAFFT

```
$ pixi add mafft
```

And don't forget to activate the environment:

```
$ pixi shell
```

Some HBV data

I put 103 HBV genomes into `/sc-projects/sc-proj-cc11-civclub/club-club/data/hbv-genomes.fasta`.

You can probably get to this as `~/data/hbv-genomes.fasta` via the symlink in your home directory.

Try this

```
$ mkdir ~/msa-class
$ cd ~/msa-class
$ mafft ~/data/hbv-genomes.fasta > hbv-genomes-aligned.fasta
```

On a compute node with 32Gb of RAM this took under three seconds.

Options you could add

- `--out` - put the output into a given file (instead of standard output).
- `--preservecase` - keep sequences in the same case as the input file.
- `--anysymbol` - allow odd characters in sequences (treat them as `N`).
- `--thread` - use multiple threads to increase speed.
- `--reorder` - make the output order match the input order.
- `--quiet` - don't print progress information.

How MAFFT works

Three steps always run. Then there's an optional fourth.

- Compute pairwise distances.
- Use the distances to make a "guide tree".
- Make a progressive alignment (starting from tips) from the tree.
- Optional: iterative refinement.

MAFFT: Pairwise alignment

This is used to get distances between pairs of sequences to make a guide tree. The "tree" is really just the result of distance-based clustering.

Full pairwise alignment can be slow.

- Fast methods count shared 6-mers (by default) in each pair. This is cheap but still an $O(N^2)$ approach
- Much slower: run a full (dynamic-programming) alignment of every pair.
- PartTree picks a few representative sequences, assigns the other sequences to their nearest representative, and recurses inside each group, so it doesn't needs the full $N \times N$ matrix. This is $O(N \log N)$.

MAFFT: Guide tree

MAFFT uses the guide tree to determine the order in which it progressively aligns sequences (or groups of sequences).

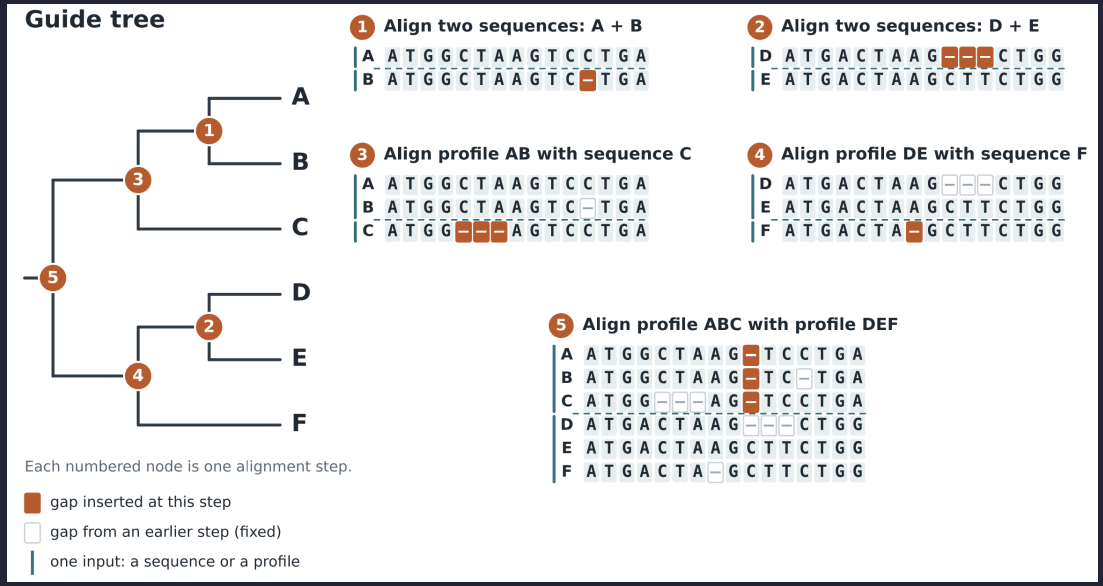
Here's how it makes the tree:

- Distances are clustered to make a rooted tree. This fixes the order in which sequences and groups are merged. Sequence weighting keeps many near-identical sequences from dominating.
- In FFT-NS-2 (the default), the first tree from k-mer distances is rough. MAFFT aligns using it, recomputes distances from that alignment, builds a better tree, and aligns again.

MAFFT: Progressive alignment

Working from the leaves of the guide tree to the root, MAFFT merges sequences into groups and groups into larger groups. Each merge is a group-to-group (profile) dynamic-programming alignment. A column pair's score is the weighted average of the substitution scores between all residue pairs across the two groups.

This is where the FFT comes in. MAFFT doesn't run dynamic programming over the full length right away. It converts each residue into numbers: for amino acids, their volume and polarity; for nucleotides, one value per base. It then uses the fast Fourier transform to compute the correlation between the two profiles at every possible offset. Peaks in the correlation mark offsets at which homologous segments line up. MAFFT locates those segments and runs dynamic programming only between and around them, which is much cheaper for long sequences. --nofft (the NW-NS methods) skips this shortcut and uses plain dynamic programming throughout.



MAFFT: Optional iterative refinement

Progressive methods have a weakness: gaps placed in early merges are fixed for the rest of the run.

Refinement goes back to correct early mistakes. MAFFT repeatedly cuts one branch of the guide tree, which splits the alignment into two sub-alignments. It removes columns that are entirely gaps, realigns the two parts against each other, and keeps the new alignment only if the objective score improves. It cycles through the branches until nothing improves or `--maxiterate` is reached.

MAFFT: other useful options

Options for adding new sequences to an existing alignment instead of realigning everything:

- `--add`
- `--addfragments`
- `--addprofile`
- `--addlong`

Maintaining the length of a pre-existing alignment:

- `--keeplength`

Letting MAFFT to decide which approach to take:

- `--auto`

MAFFT: How `--auto` works

You can run with `--auto` if you want MAFFT to decide which approach to take. `--auto` measures the number of sequences and their length then takes the first rule that applies:

- < 100 sequences and length < 3,000: L-INS-i
- < 200 sequences and length < 1,000: L-INS-i with 2 iterations
- < 500 sequences and length < 10,000: FFT-NS-i, 2 iterations
- < 20,000 sequences: FFT-NS-2
- < 100,000 sequences: FFT-NS-2 with a memory-saving tree
- < 200,000 sequences: FFT-NS-1 with a memory-saving tree
- Else: PartTree variants

Taking a step back

Pairwise sequence alignment

- Is a solved problem.
- Can be done on full sequences or locally
- There are well-known algorithms for both

Global — Needleman–Wunsch (1970)



Every residue of both sequences is placed; unmatched stretches become gaps.

Aligns the full length of both sequences. Right when the sequences are homologous end to end, e.g. orthologous genes.

Local — Smith–Waterman (1981)



Only the shared segment (dark) is aligned; flanking sequence is not scored.

Finds the best-matching segments and ignores the rest. Right for shared domains, or a short query inside a long subject.

Amino acid alignments

Many packages can make protein alignments. This can be much better / more appropriate in some circumstances.

Codon-aware alignments

A nucleotide aligner doesn’t know the reading frame

It scores bases one at a time, so coding sequences go wrong in three ways

1 Indels split into pieces

IN FRAME

```
ATG GCA AAG GTC TGA
ATG GCA --- GTC TGA
```

NUCLEOTIDE ALIGNER

```
ATG GCA AAG GTC TGA
ATG GC- A-- GTC TGA
```

One 3-nt deletion becomes a 1-nt and a 2-nt gap. The A of sequence 2’s GCA codon ends up in the next codon.

2 Gaps off codon boundaries

IN FRAME

```
ATG GCT GCA AAA
ATG --- GCA AAA
```

EQUALLY GOOD TO THE ALIGNER

```
ATG GCT GCA AAA
ATG G-- -CA AAA
```

The 3-nt gap scores the same at three positions; only one is in frame. Read in triplets, the other two give codons like G-- and -CA.

3 The genetic code is invisible

DNA: 2 OF 9 IDENTICAL (22%)

```
CTG AGC CGT
TTA TCT AGA
```

PROTEIN: 3 OF 3 IDENTICAL

```
Leu Ser Arg
Leu Ser Arg
```

Synonymous changes make related sequences look unrelated, so the aligner may shift or gap them to chase a few base matches.

Schematic examples. In panels 1 and 2, sequence 2 has lost one codon relative to sequence 1.

Upcoming classes

- Tree making.
- The dark-matter tools.
- de novo assembly.
- Protein-level matching.